



Machine Learning Approach to Predicting Academic Dropout

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Course: CSCI 3052U
CRN: 46506

Abstract

Universities lose millions of dollars every year to student dropout, yet most still lack a formal system for catching students who are struggling academically. Advisors mostly rely on informal signals, like noticing a low GPA, and by the time they notice it, the student has often been behind for a while. Machine learning research increasingly suggests this predictive gap can be closed, and datasets already exist with the demographic, socioeconomic, and academic information needed to try. The problem is that these datasets rarely label outcomes cleanly as dropout, enrolled, or graduate.

This project applies machine learning to one such dataset: the “Predict Students' Dropout and Academic Success” dataset from the UCI repository, which covers roughly 4,400 students with demographic and socioeconomic data plus first- and second-semester grades. Each student is labeled Dropout, Enrolled, or Graduate, so we treat this as a multi-class classification problem. We'll compare several approaches, including a naive-majority baseline, logistic regression, k-nearest neighbors, decision trees, and ensemble methods, testing all of them on a held-out sample using macro-average F1 as the evaluation metric. We'll also run a subgroup error analysis to check that students are treated consistently regardless of background factors like prior qualifications or parental education. The goal is to show whether a systematic, evidence-based early-warning tool is feasible as a supplement to how advisors currently catch at-risk students.

Research Background

University dropout remains common in higher education. Many institutions report that first-year dropout rates can range anywhere from 20-30%, which leads to real financial and personal costs for both students and the institution. This project's task is to predict, from information already on record early in a student's time at the university, whether that student is likely to drop out, stay enrolled, or graduate.

Most universities still catch at-risk students informally. These methods include an instructor or advisor noticing missed classes, or a GPA cutoff that gets applied once a student has already started struggling. This approach leads to catching the problem late rather than early and rarely considers the demographic, socioeconomic, and early-performance data that enrollment offices usually already have.

Automated dropout prediction is not new to educational data mining, but a small team rarely gets to build, evaluate, and interpret a full pipeline end-to-end on a public dataset within a single course term. That is the gap this project fills, comparing several course-taught model families fairly rather than reporting one best-tuned result.

Methodology

Data: We plan on using the “Predict students’ Dropout and Academic Success” dataset from the UCI Machine Learning Repository (Realinho et al., 2021). This repository contains about 4,400 anonymous students’ data (i.e., approximately 30+ features that relate to demographic information, grade in admissions, socioeconomic status, and academic performance of the students in their first and second semester courses). The data we’ll be using is labelled by one of three outcomes (dropout/enrolled/graduate). The dataset is available to researchers, educators, and students for free and does not require collecting additional data or obtaining human subject approvals.

Methods: We will develop a naive majority-class baseline. In addition, we will test a linear course-taught baseline (logistic regression), k nearest neighbours, decision tree, and ensemble approach (random forest or gradient boosting) with the same preprocessed features and data split. The data will be split into train-validation-test set with stratification (approximately 70%, 15%, 15%), where the validation set will be used to choose the best model and tuning of the parameters. Also, the test set will be touched only once during the evaluation of the overall results.

Significance

The main contribution here is a reproducible benchmark: we want to see whether enrollment and first-semester data can actually predict dropout risk. We’re comparing a naive baseline against several methods from this course under one fixed evaluation protocol, rather than just reporting whichever run happened to look best.

Academic advising and retention offices could use this evidence to decide whether a systematic early-warning tool is worth building out. If one ever gets adopted, students benefit indirectly through earlier, more consistent outreach. We get something out of it too: real practice comparing model families under a shared, documented evaluation protocol instead of one-off experiments.

Right now, dropout identification is mostly reactive: an advisor notices a problem informally, practices vary by department, and by the time anyone flags a student, they’re often already struggling. This project points toward a proactive alternative, built on data a registrar’s office already collects at enrollment.

We’ll call the project a success if at least one trained model beats the naive baseline by a meaningful margin on macro-F1, the full pipeline reproduces from the repository with documented splits and seeds, and the subgroup error analysis doesn’t turn up any severe, unexplained performance gap across attributes like prior qualification or parental education. A positive result could be extended with newer enrollment cohorts, more features, or a fuller fairness audit. A negative or mixed result is still useful; it tells an advising office what doesn’t work, and that could shape future course projects or lab work on the same dataset.

SMART Timeline

Milestone 1 – Proposal & Team Contract (Sept 18, 2026)

Finalize the project proposal, submit the team contract, and confirm role assignments.

Milestone 2 – Dataset & Problem Approval (Oct 2, 2026)

Confirm dataset license/access, complete initial EDA, document class balance and missingness, and fix the train/val/test split and primary metric.

Milestone 3 – Baseline Reproduction (Oct 19, 2026)

Working data pipeline, naive benchmark, and course-level baseline implemented and documented with a baseline results table. Complete Start/Stop/Continue plan.

Milestone 4 – Progress Presentation (Oct 30, 2026)

Preliminary results from at least two additional models (k-NN, decision tree, and/or ensemble) and revised timeline based on progress to date.

Milestone 5 – Final Experiments & Analysis (Nov 20, 2026)

Complete fair comparison across all models, tuning procedure, uncertainty/sensitivity analysis, and draft results section.

Milestone 6 – Final Report, Repository & Demo (Dec 4, 2026)

Final 6-8 page report, tagged reproducible repository release, and final presentation/demo.

Milestone 7 – Individual Reflection & Oral Defence (Dec 7, 2026)

Submit individual reflections and oral defence.

References

- Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32. <https://doi.org/10.1023/A:1010933404324>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.